

2-①

問1 C_5H_{10}

問2 A: $CH_2 = \underset{\substack{| \\ CH_3}}{C} - CH_2 - CH_3$

B: $CH_2 = CH - \underset{\substack{| \\ CH_3}}{CH} - CH_3$

C: $CH_3 - CH = CH - CH_2 - CH_3$

問3 C: $\begin{array}{c} CH_3 \quad \quad CH_2 - CH_3 \\ \quad \diagdown \quad \diagup \\ \quad C = C \\ \quad \diagup \quad \diagdown \\ H \quad \quad H \end{array}$

2-②

問1 C_6H_{12}

問2 A: $CH_3 - CH = \underset{\substack{| \\ CH_3}}{C} - CH_2 - CH_3$ B: $CH_3 - \underset{\substack{| \\ CH_3}}{C} = \underset{\substack{| \\ CH_3}}{C} - CH_3$

C: $CH_3 - CH_2 - CH = CH - CH_2 - CH_3$ D: $CH_2 = \underset{\substack{| \\ CH_2 - CH_3}}{C} - CH_2 - CH_3$

問3 A > C

2-③

問1 A $CH_2 = CH - CH_2 - CH_3$ 1-ブテン

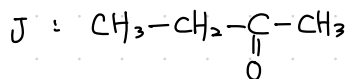
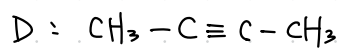
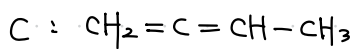
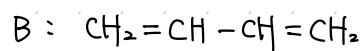
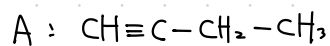
B $CH_3 - CH = CH - CH_3$ 2-ブテン

C $CH_3 - CH_2 - CH_2 - CH_3$ ブタン

問2 D: 2 E: 3 (メソ体あり)

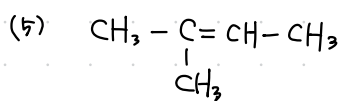
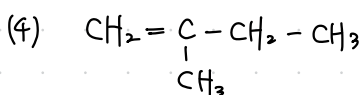
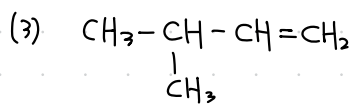
問3 $\begin{array}{cc} CH_2 - CH_2 & CH_2 \\ | & / \quad \backslash \\ CH_2 - CH_2 & CH_2 - CH_2 - CH_3 \end{array}$

2 - 4



2 - 5

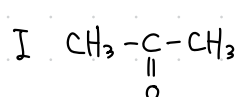
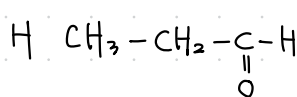
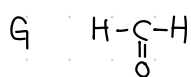
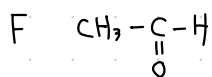
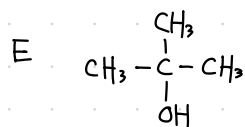
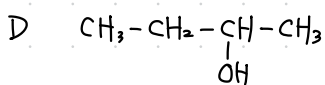
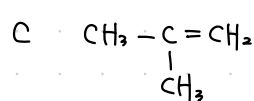
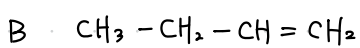
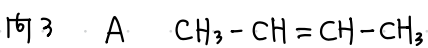
(1) 5 (2) 1



2 - 6

161 102

162 C_4H_8



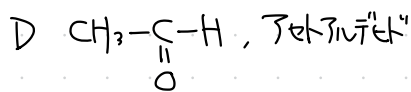
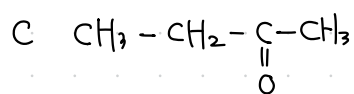
164 F

2 - 7

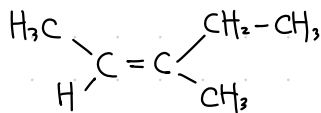
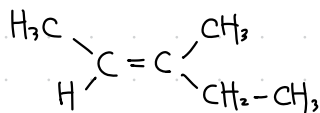
問1 略

問2 C C_4H_8O D C_2H_4O

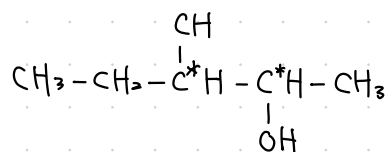
問3



問4.



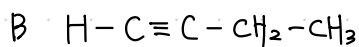
問5



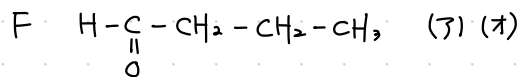
2 - 8

問1 アセチレン, γ -ブチレン

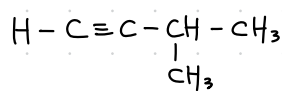
問2 (1) A $CH_3-C \equiv C-CH_3$



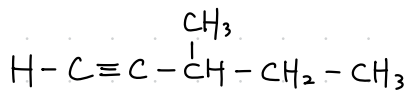
(2) E $CH_3-\overset{\overset{O}{\parallel}}{C}-CH_2-CH_3$ (イ)



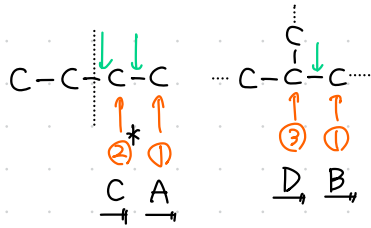
問3



問4.

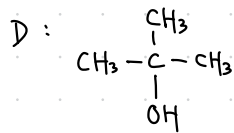
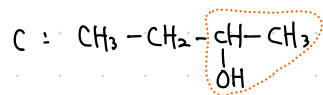
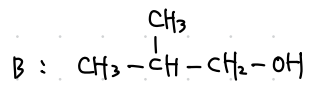
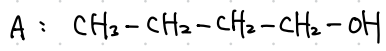


3-11 $C_4H_{10}O$ $U=1$ $\left\{ \frac{\text{アルコール}}{\text{エーテル}} \right.$



アルコール $A^{\circ} \quad B^{\circ} \quad C^* \quad D^{\circ}$
 O_x $\underline{E} \quad \underline{F} \quad G \quad X$
 アルコール アルコール
 b.p. $A > B$

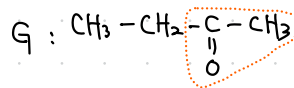
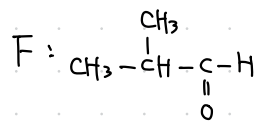
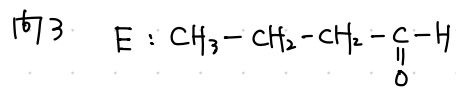
問1



問2

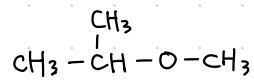
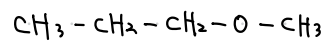
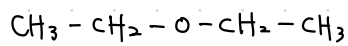
ヒドロキシ基を持ち、
分子間での水素結合を形成するため。

問3

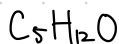


C, G

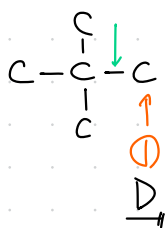
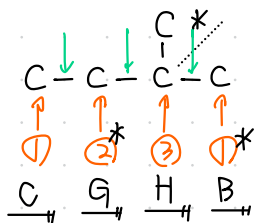
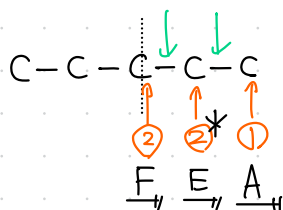
問4



3 - 2

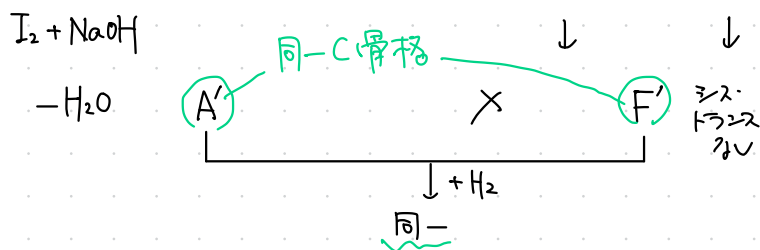


$$U=0 \quad \left\{ \begin{array}{l} \gamma_{\mu\nu} = -\gamma_{\nu\mu} \\ I = -\bar{I} \end{array} \right.$$



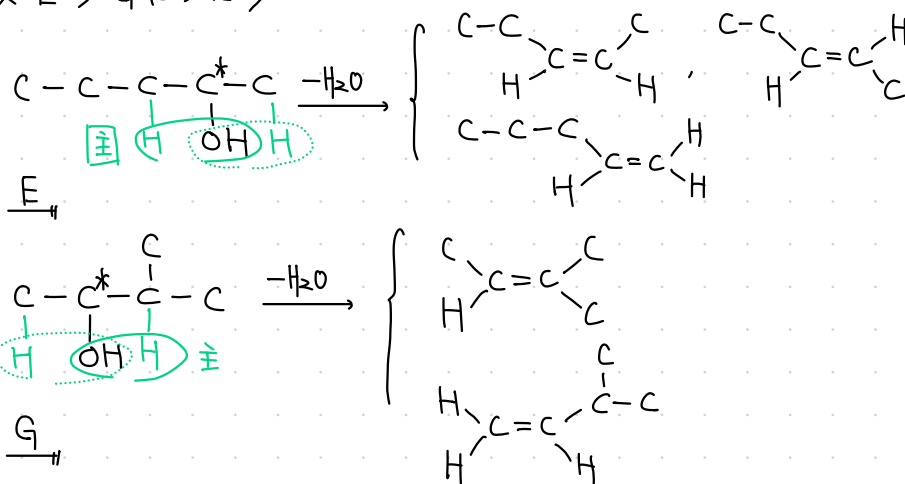
アルコール ① A ① B^{*} ① C ① D ② E^{*} ② F ③ G^{*} ③ H

Ox { アルデヒド } { ケトン } X



$\left. \begin{array}{l} \text{「脱水した} \sim \text{」} \\ \text{「H}_2\text{付加した} \sim \text{」} \end{array} \right\} \text{同一化合物}$
 $\longrightarrow \text{同一C骨格}$

$\langle 2 \text{級 } E^*, G^* | \Rightarrow WZ \rangle$

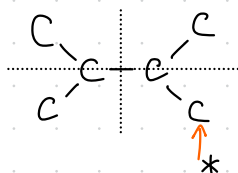
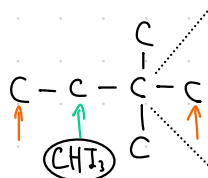
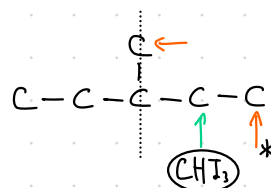
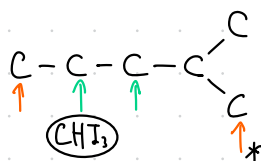
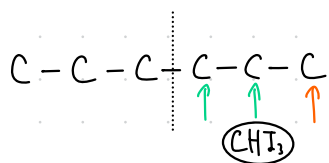


(2) $\underline{6}$ (立体异构: 7)

3 - ③

← : = O (アルデヒド)
← : = O (ケト>)

問1

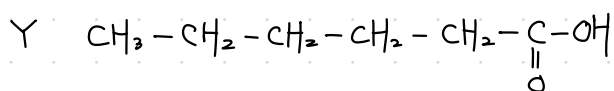
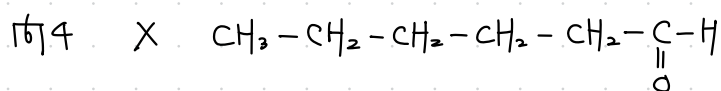
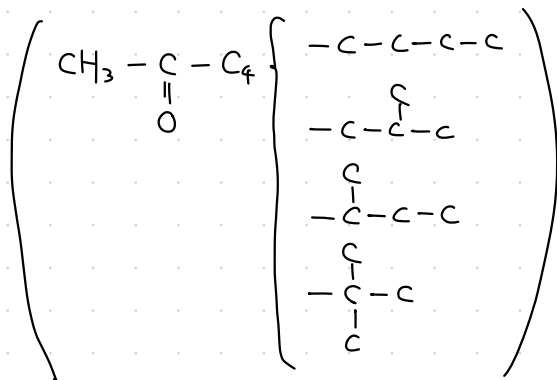


(a) 8

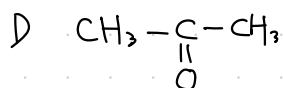
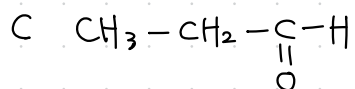
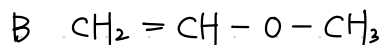
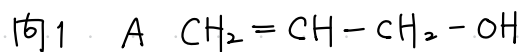
(b) 6

問2. 3

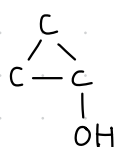
問3. 4



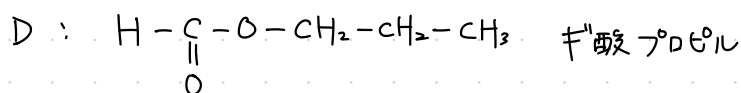
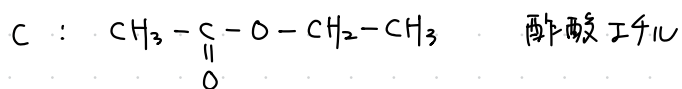
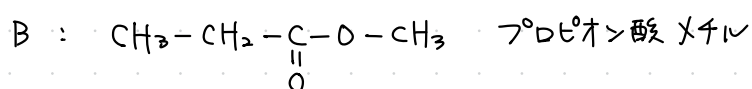
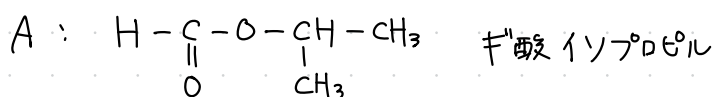
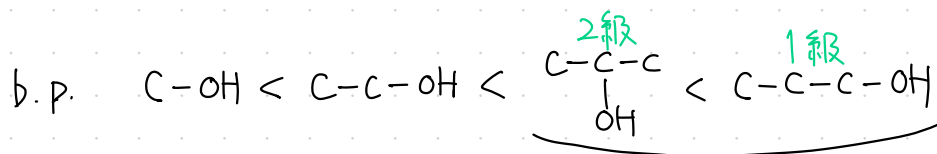
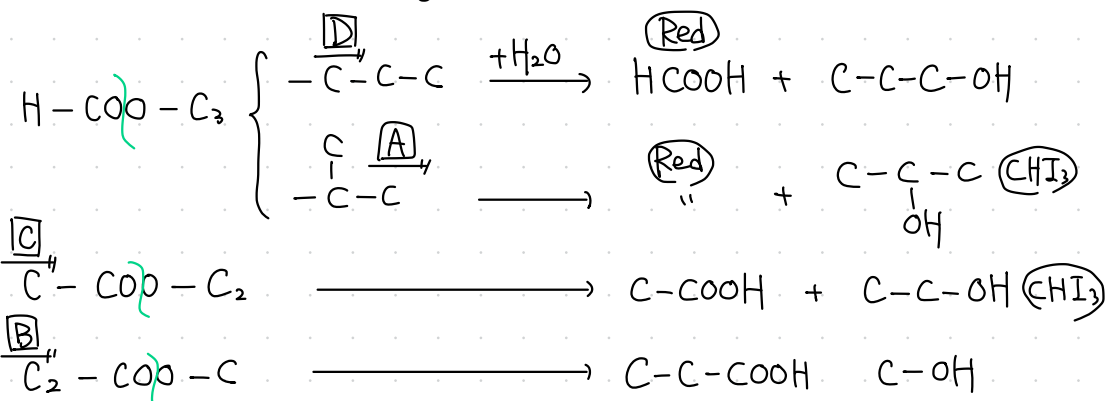
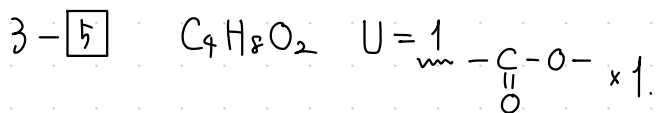
3 - ④



問2



3



3-6

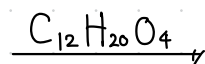
$$\text{例1} \quad w_C = 264 \text{ g} \times \frac{C(12)}{CO_2(44)} = 72 \text{ g}$$

$$w_H = 90 \text{ g} \times \frac{2H(2)}{H_2O(18)} = 10 \text{ g}$$

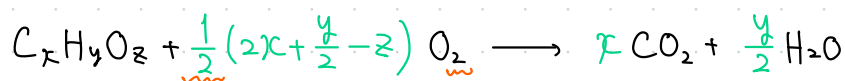
$$w_O = 114 - (72 + 10) = 32 \text{ g}$$

$$C : H : O = \frac{72 \text{ g}}{12 \text{ g/mol}} : \frac{10 \text{ g}}{1.0 \text{ g/mol}} : \frac{32 \text{ g}}{16 \text{ g/mol}} = 3 : 5 : 1$$

$$(C_3H_5O)_n = 57n = 228 \quad n = 4$$



<別解> 燃烧例, 分子量既知の時



$$\begin{matrix} C_xH_yO_z : CO_2 = 1 : x \text{ mol} \\ (288) \quad \quad (44) \end{matrix}$$

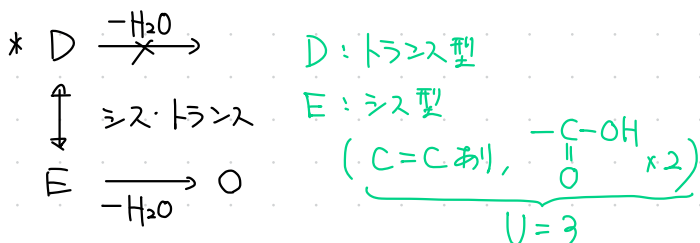
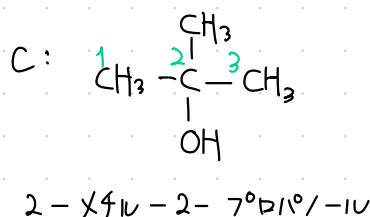
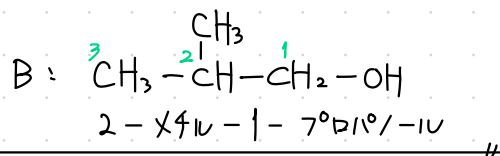
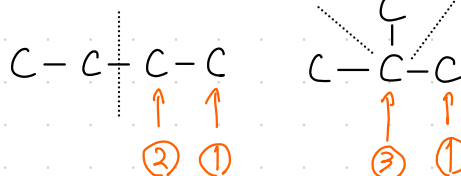
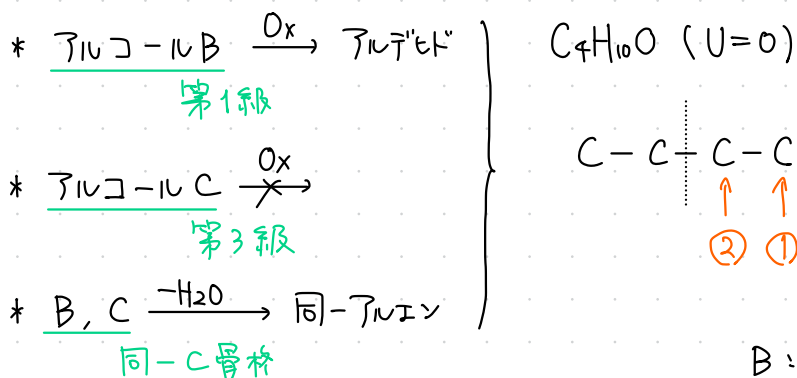
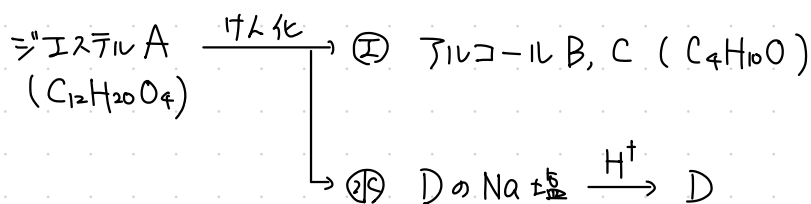
$$\frac{114 \text{ g}}{288 \text{ g/mol}} \times x = \frac{264 \text{ g}}{44 \text{ g/mol}} \times 1 \quad \therefore x = 12$$

$$\begin{matrix} C_xH_yO_z : H_2O = 1 : \frac{y}{2} \text{ mol} \\ (288) \end{matrix}$$

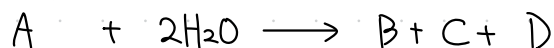
$$\frac{114 \text{ g}}{288 \text{ g/mol}} \times \frac{y}{2} = \frac{90 \text{ g}}{18 \text{ g/mol}} \times 1 \quad \therefore y = 20$$

$$C_{12}H_{20}O_z = 164 + 16z = 228 \quad \therefore z = 4 \quad \underline{C_{12}H_{20}O_4}$$

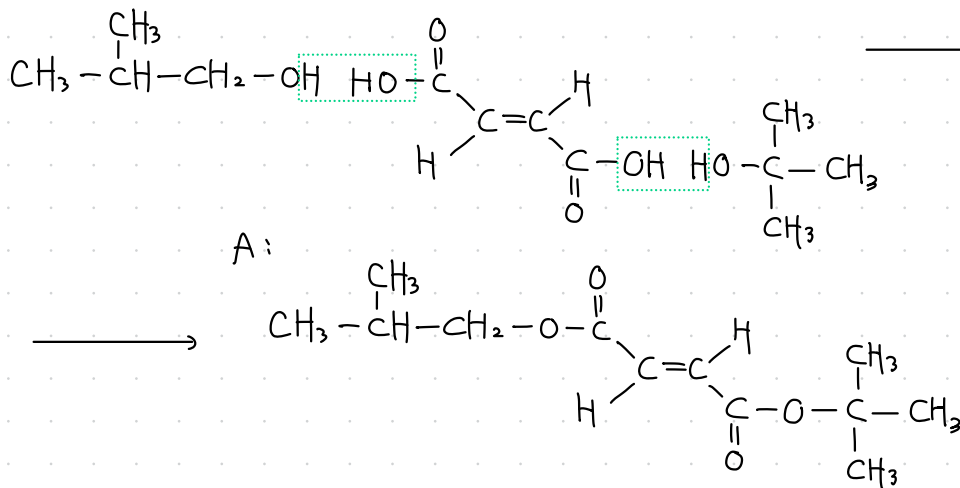
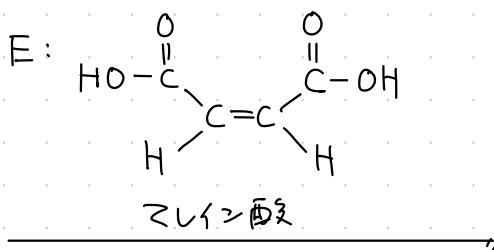
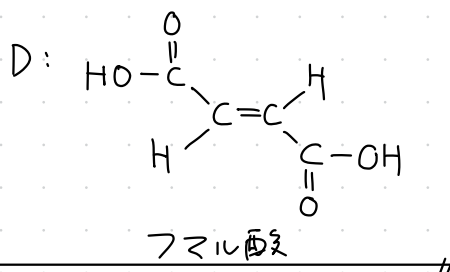
問2. A: $C_{12}H_{20}O_4$ $U = \frac{2 \times 12 + 2 - 20}{2} = 3$ C=C $\times 1$ + $-\overset{\overset{O}{\parallel}}{C}-O- \times 2$



ジエステル



$\therefore D = A + 2H_2O - (B + C)$
 $= C_{12}H_{20}O_4 + 2H_2O - 2 \times C_4H_{10}O$
 $= C_4H_4O_4$ (114)
 ($U=3$)



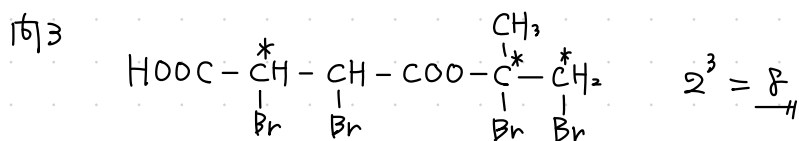
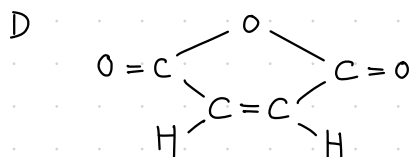
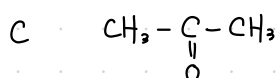
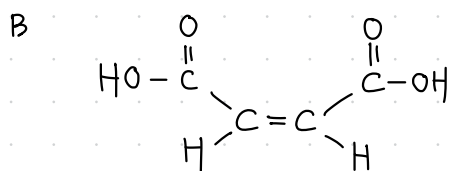
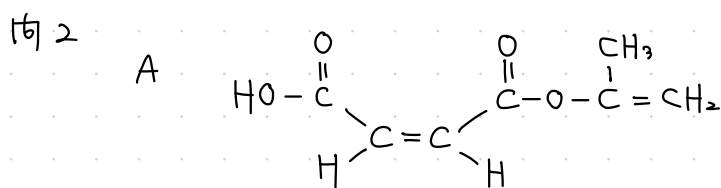
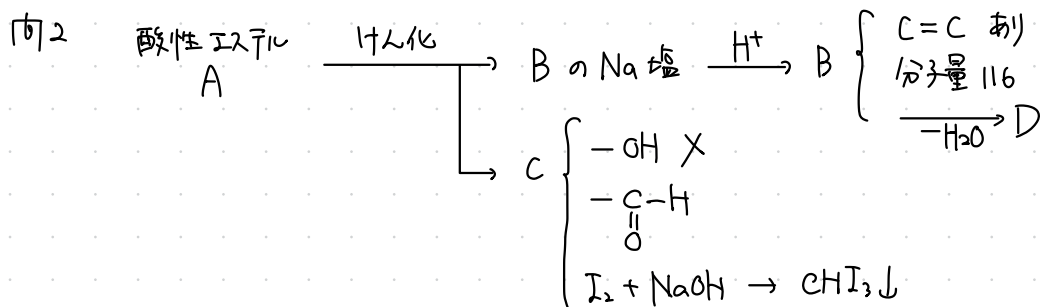
3 - [7]

問1 $C_7H_8O_4$

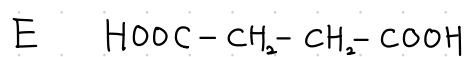
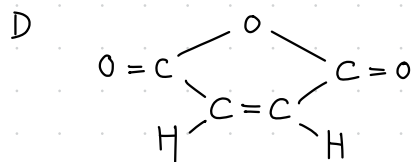
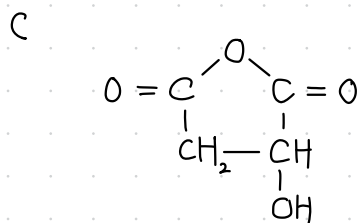
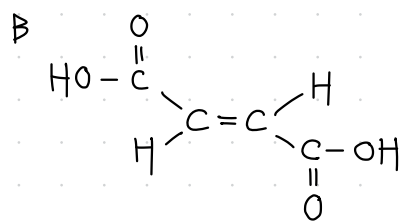
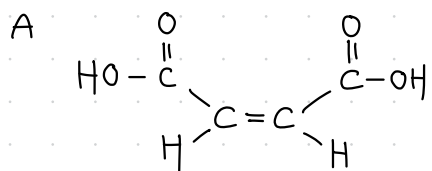
$$C:H:O = \frac{53.8}{12} : \frac{5.1}{1.0} : \frac{41.1}{16} = 4.48 : 5.1 : \frac{2.56}{\text{①}}$$

$$\div 1.75 : 1.99 : 1 = 7:8:4$$

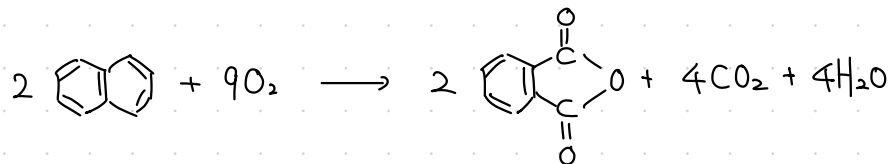
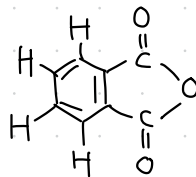
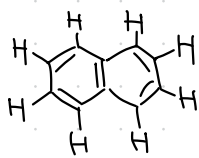
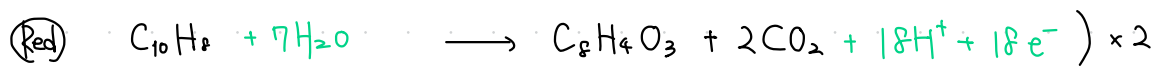
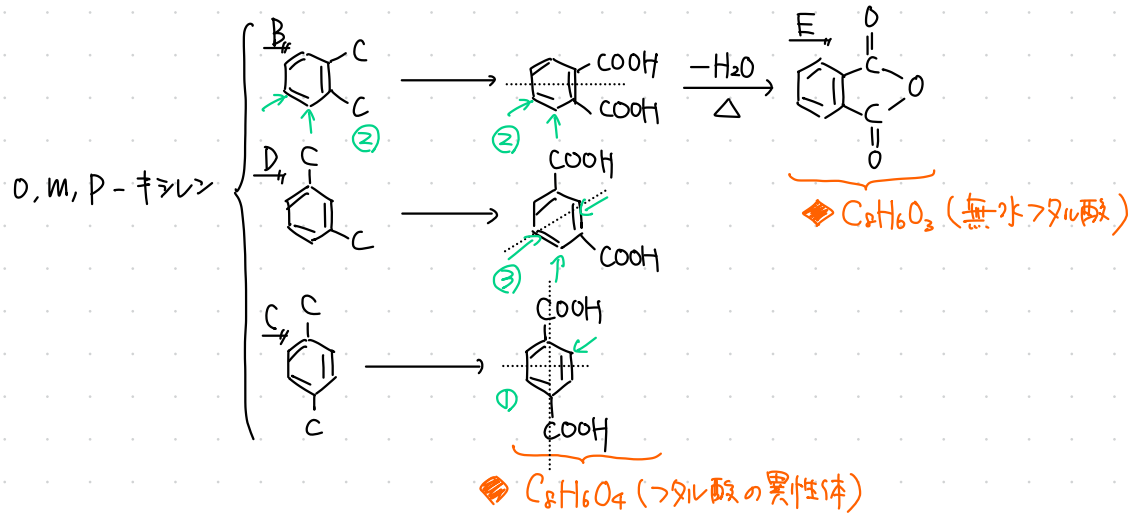
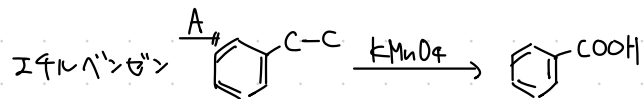
$$130 \leq (C_7H_8O_4)_n \leq 170 \quad \therefore n=1. \quad \underline{C_7H_8O_4} \quad (U=4)$$



3 - 8



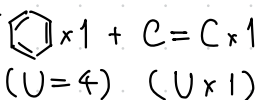
4-① C_8H_{10} $U = 4$  $\times 1$



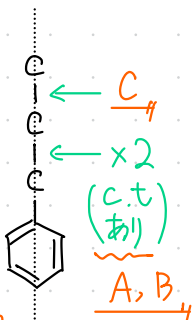
4-2

C_9H_{10}

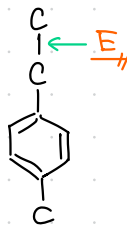
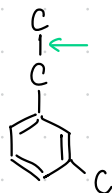
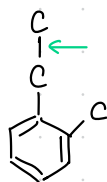
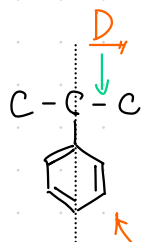
$U = 5$



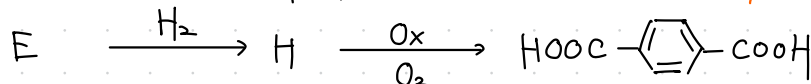
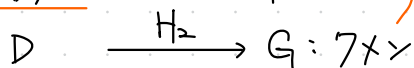
内2.



< 1置換体 >

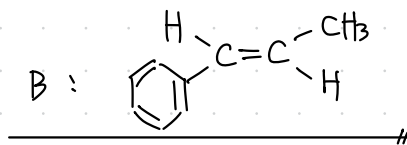
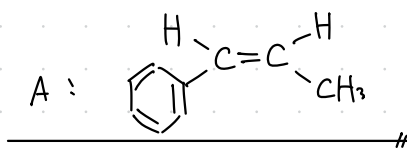


< 2置換体 >



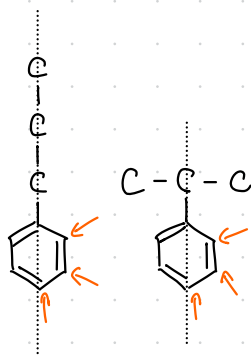
A, B: シス・トランス異性体

融点 $A < B$
シス トランス



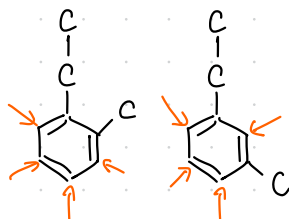
内1. 8

内3. 22



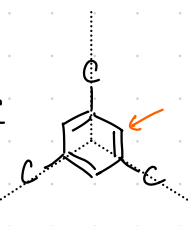
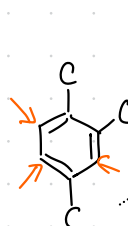
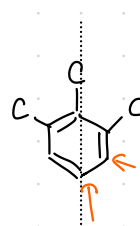
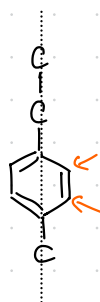
< 1置換体 >

Cl置換: 6



< 2置換体 >

Cl置換: 10



< 3置換体 >

Cl置換: 6

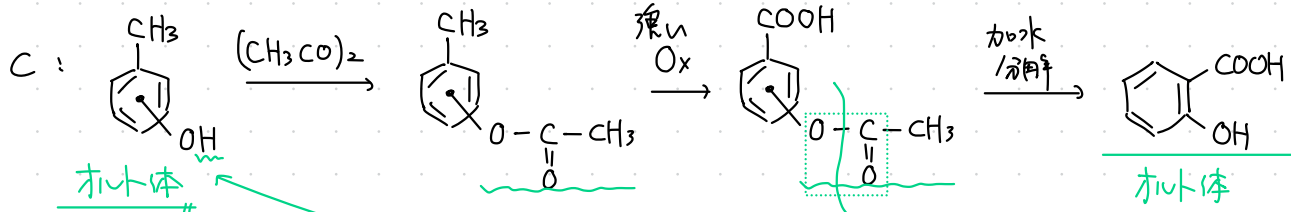
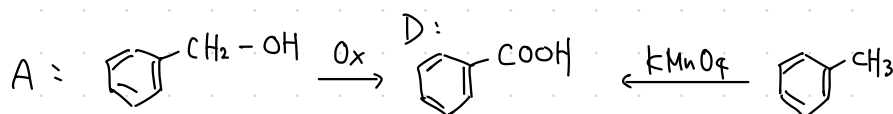
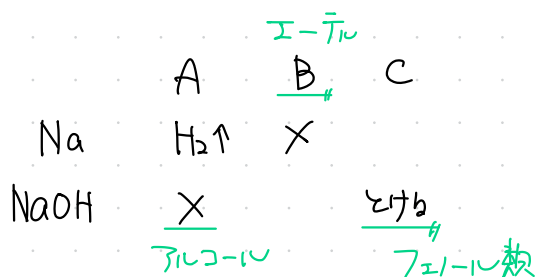
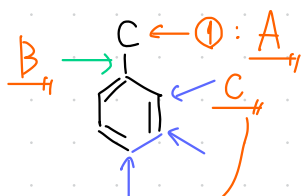
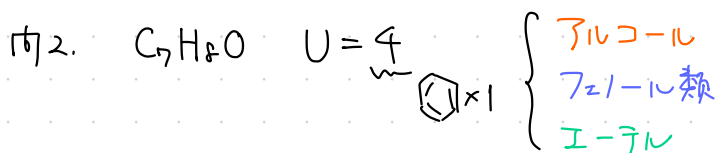
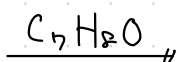
4-③

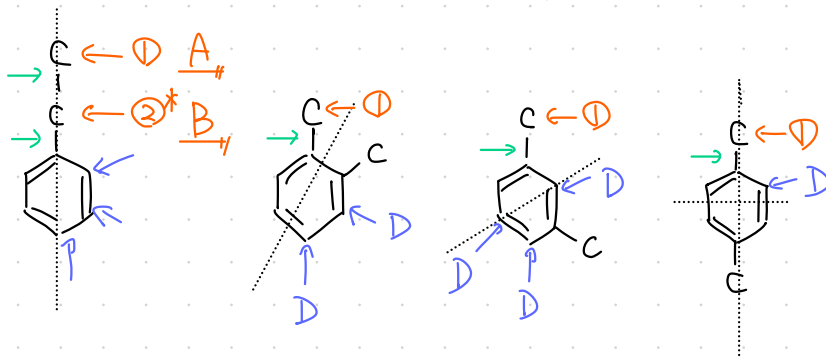
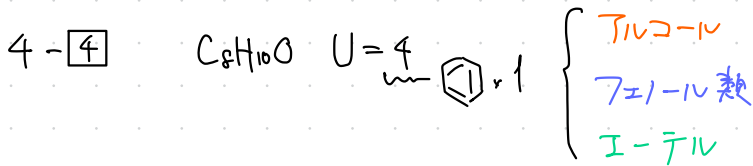


条件より $l : \frac{m}{2} = 7 : 4 \quad \therefore (l, m) = (7, 8)$

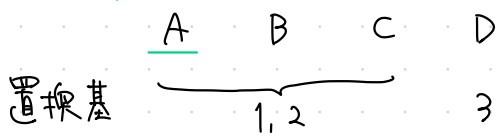
$\therefore (C_7H_8O_n)_k = (92 + 16n)k = 108$

$\therefore (n, k) = (1, 1)$

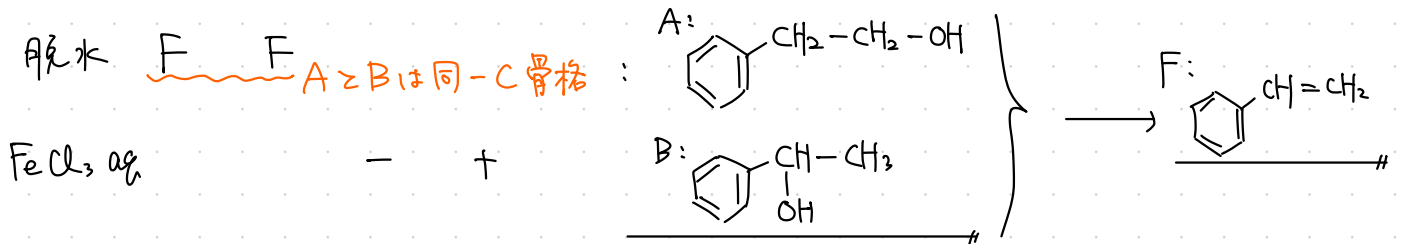




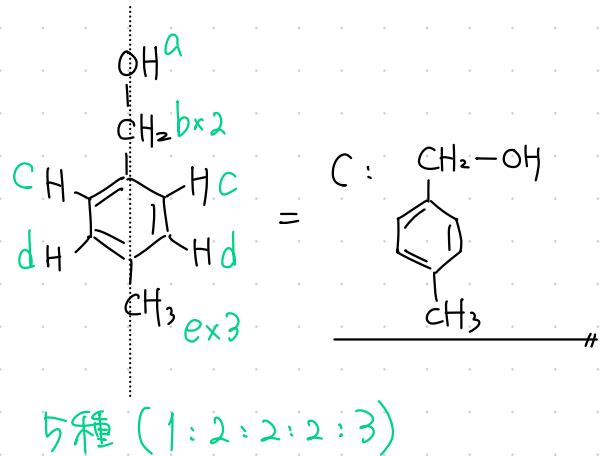
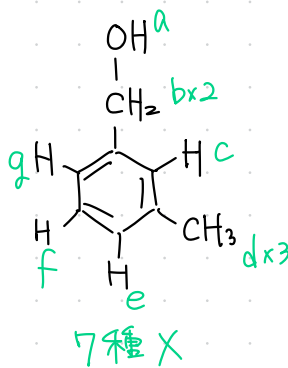
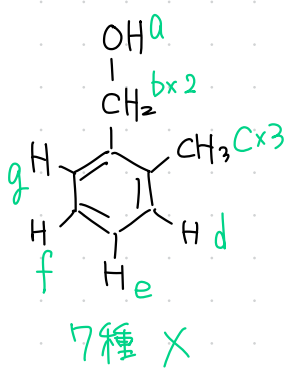
第1級
アルコール



おだやみ
Ox $\underline{E}$ アルデヒド



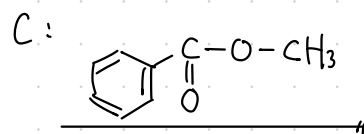
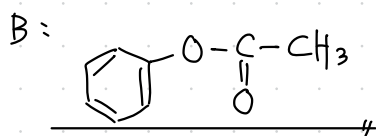
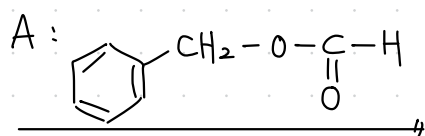
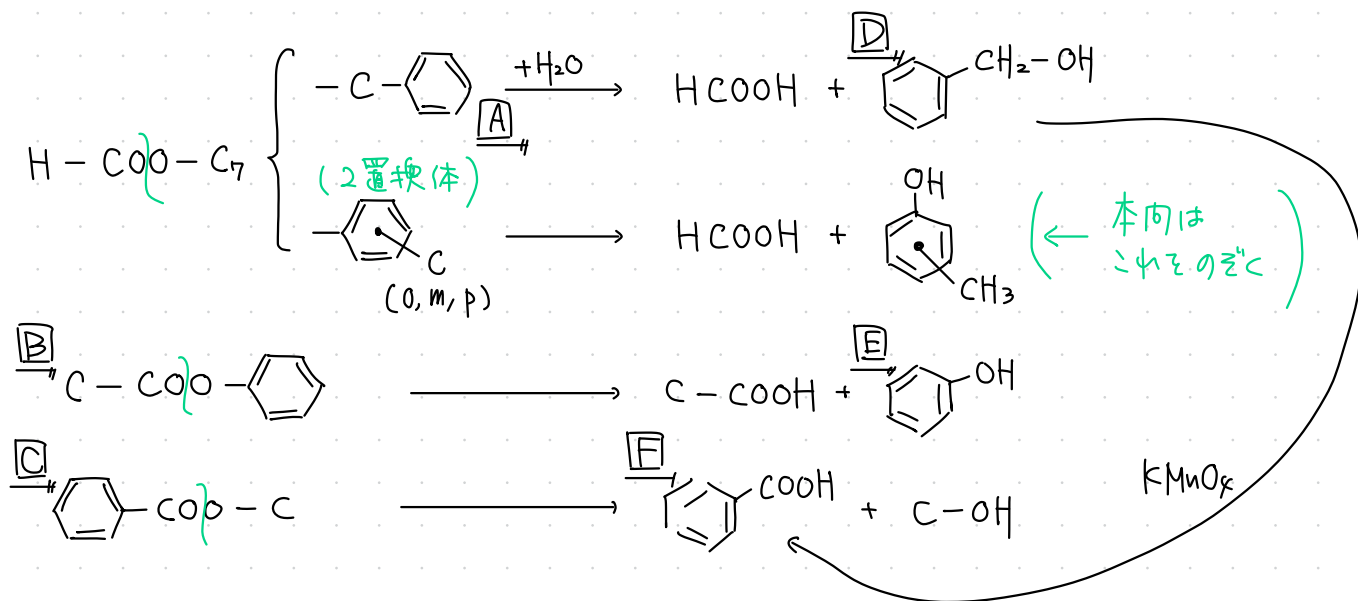
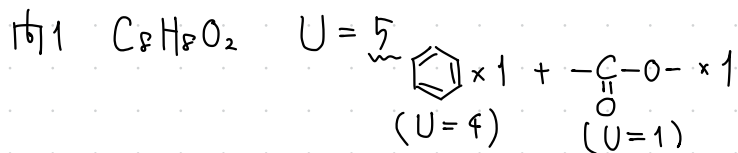
C : 1 or 2 置換体のアルコール



内2

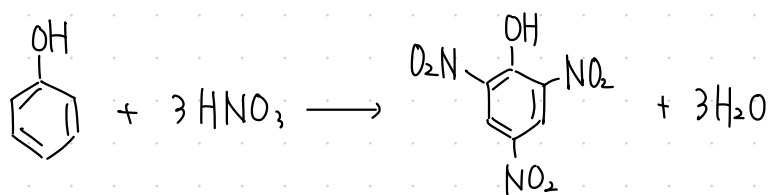
D: フェニル類 (3 置換体), 6

4-7



162. E

163



ヒョウリ=酸

2, 4, 6 - トリトロフエノール